

NETFLIX

CONTENT CATALOG ANALYSIS

An Exploratory Data Analysis of 8,807 Titles Across Python, SQL & Power BI

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1. Executive Summary

This report presents a comprehensive exploratory data analysis (EDA) of the Netflix global content catalog, comprising 8,807 titles spanning movies and TV shows released between 1925 and 2021. The analysis was conducted using a multi-tool workflow — Python (Pandas, Matplotlib) for exploratory analysis, PostgreSQL for structured querying, and Power BI for interactive dashboarding — reflecting an end-to-end analytics pipeline from raw data to business-ready insight.

The objective is to understand Netflix's content strategy across four lenses: content composition (movies vs. TV shows), geographic distribution of production, genre and audience-rating trends, and the evolution of the catalog over time. Findings are intended to support content acquisition, regional investment, and catalog-expansion decisions.

8,807 Total Titles	69.6% Movies Share	125 Countries Represented	2019 Peak Content-Addition Year
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2. Methodology & Data Sources

The underlying dataset (`netflix_titles`) contains 8,807 records with 12 attributes, including title, type, director, cast, country, date added, release year, rating, duration, and genre classification ("`listed_in`"). The analysis was carried out in three complementary environments:

- Python (Jupyter Notebook): Data cleaning, exploratory analysis, and visualization using Pandas and Matplotlib across 29 structured business questions.
- PostgreSQL: The cleaned dataset was loaded into a relational database (`netflix_analysis` table) to perform set-based aggregations, window functions (`RANK`, `LAG`), and year-over-year growth calculations.
- Power BI: An interactive dashboard (`netflix_dashboard.pbix`) was built for stakeholder-facing exploration of the same metrics with filtering and drill-down capability.

Standard data-cleaning steps were applied prior to analysis, including parsing the "duration" field (stripping "min" and converting to numeric minutes for movies), exploding multi-value fields such as country, cast, director, and `listed_in` (which contain comma-separated lists) into individual rows for accurate counting, and converting `date_added` to a proper datetime type.

3. Content Composition: Movies vs. TV Shows

Movies dominate the Netflix catalog, accounting for 6,131 titles (69.6%) compared to 2,676 TV shows (30.4%) — a ratio of roughly 2.3 movies for every TV show. This suggests Netflix's library has historically been built on a film-heavy acquisition strategy, though the platform has invested increasingly in original series in recent years.

Content Mix: Movies vs. TV Shows

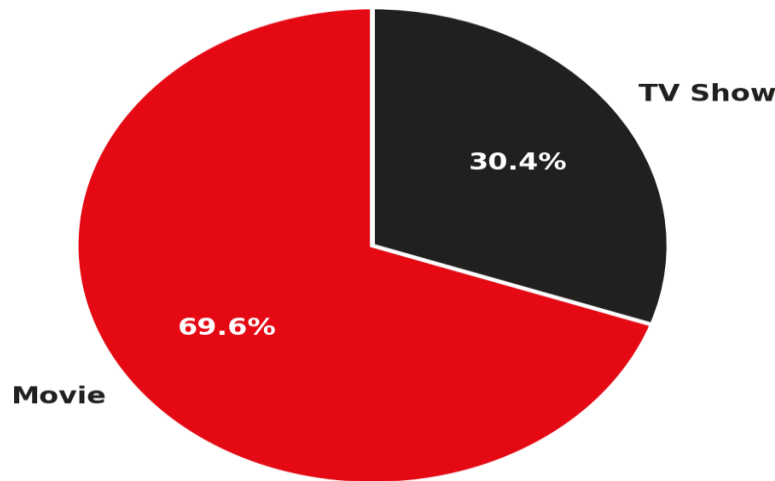


Figure 1: Proportional split of Movies vs. TV Shows in the catalog.

4. Geographic Distribution of Content

Content is sourced from over 125 countries, but production is heavily concentrated. The United States alone contributes 3,689 titles — more than the next five countries combined. India follows as the second-largest content source with 1,046 titles, reflecting Netflix's strategic investment in the South Asian market, followed by the United Kingdom, Canada, and France.

Top 10 Content-Producing Countries

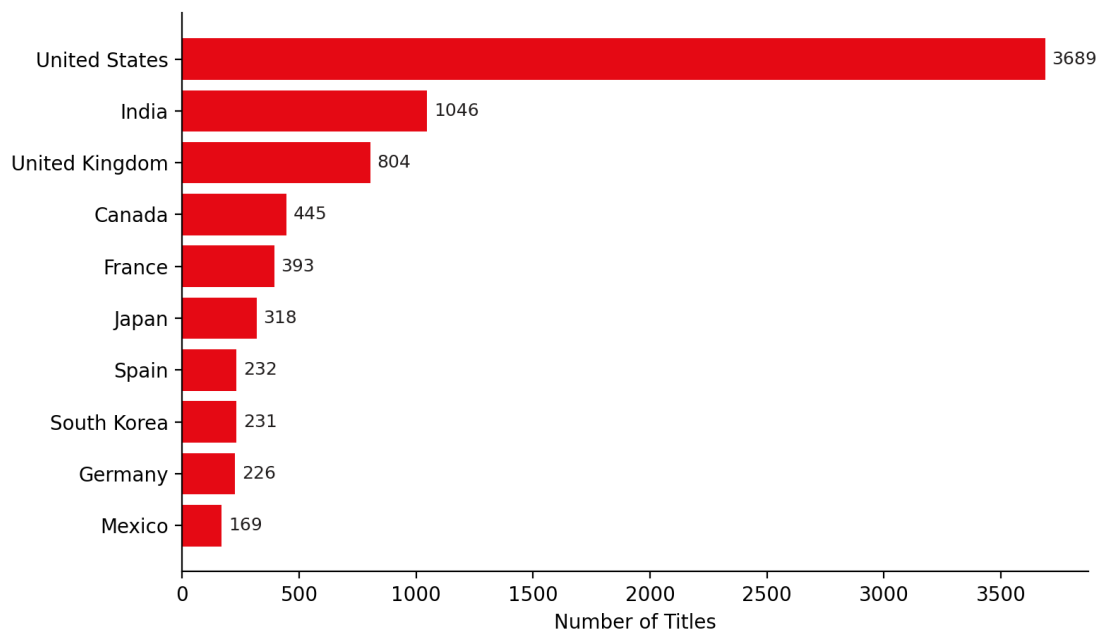


Figure 2: Top 10 content-producing countries by title count.

Notably, when content type is examined by country, several markets show a distinct skew toward TV shows over movies. South Korea (73.6% TV share) and Japan (62.6% TV share) stand out as TV-drama and anime-driven markets, while countries such as Nigeria, Turkey, and most of Latin America lean overwhelmingly toward movie content.

5. Genre & Category Trends

International Movies is the single largest genre tag (2,752 titles), followed by Dramas (2,427) and Comedies (1,674). International TV Shows (1,351) leads the TV-show genre segment. This heavy weighting toward "International" tags reflects Netflix's global-first content strategy rather than a US-centric one.

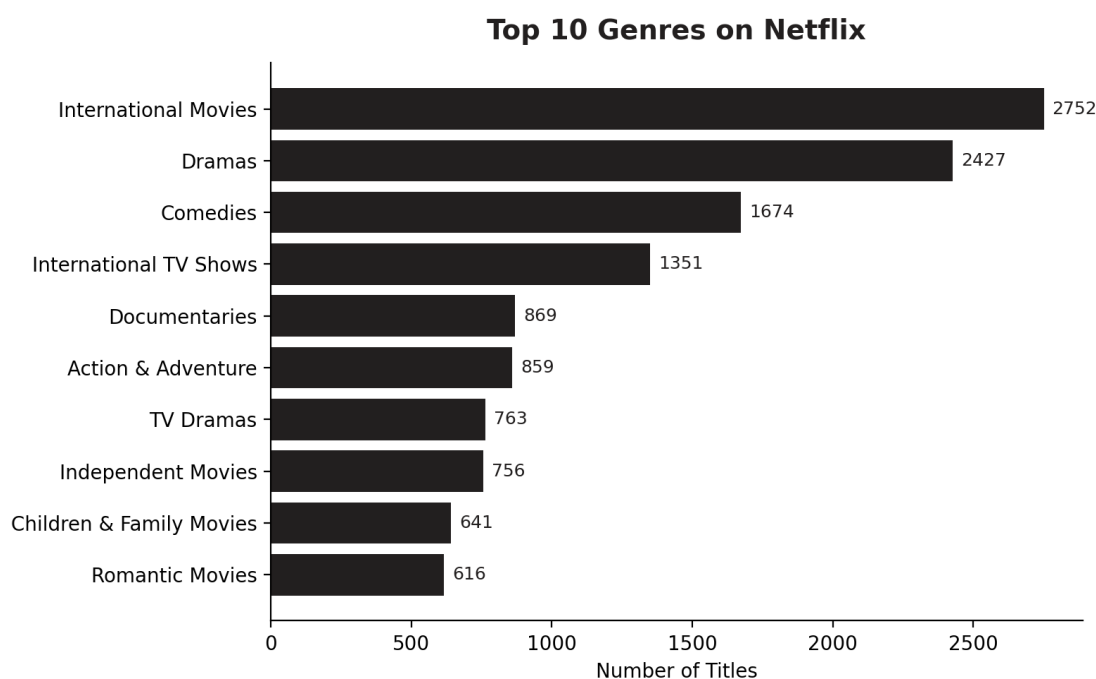


Figure 3: Top 10 genres by number of tagged titles (a title may carry multiple genre tags).

A decade-over-decade genre comparison shows Dramas, International Movies, and Documentaries experienced the steepest declines in new-release tagging in the most recent five-year window, suggesting a maturing or saturated segment, while genres tied to regional originals continue to expand.

6. Content Ratings & Audience Maturity

TV-MA (mature audience) is the most common rating, applied to 3,207 titles, followed by TV-14 (2,160) and TV-PG (863). This indicates the catalog skews toward adult and young-adult audiences rather than children's programming, though a meaningful children's and family segment exists (TV-Y, TV-Y7, PG, G — collectively over 900 titles).

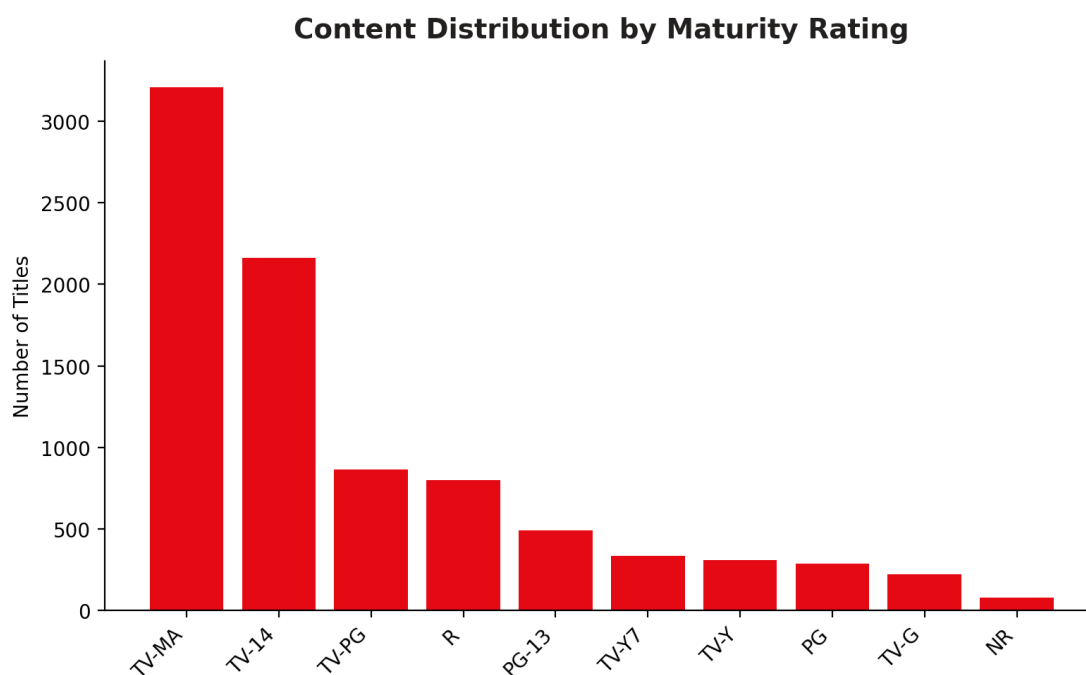


Figure 4: Distribution of titles across maturity rating categories.

7. Catalog Growth Over Time

Two distinct timelines matter for this analysis: `release_year` (when a title was originally produced) and `date_added` (when it joined the Netflix platform). By release year, content volume grew steadily through the 2010s and peaked in 2018 with 1,147 titles. By platform addition date, 2019 was the single busiest year, with 2,016 new titles added to Netflix — consistent with the platform's aggressive global-expansion phase between 2016 and 2020.

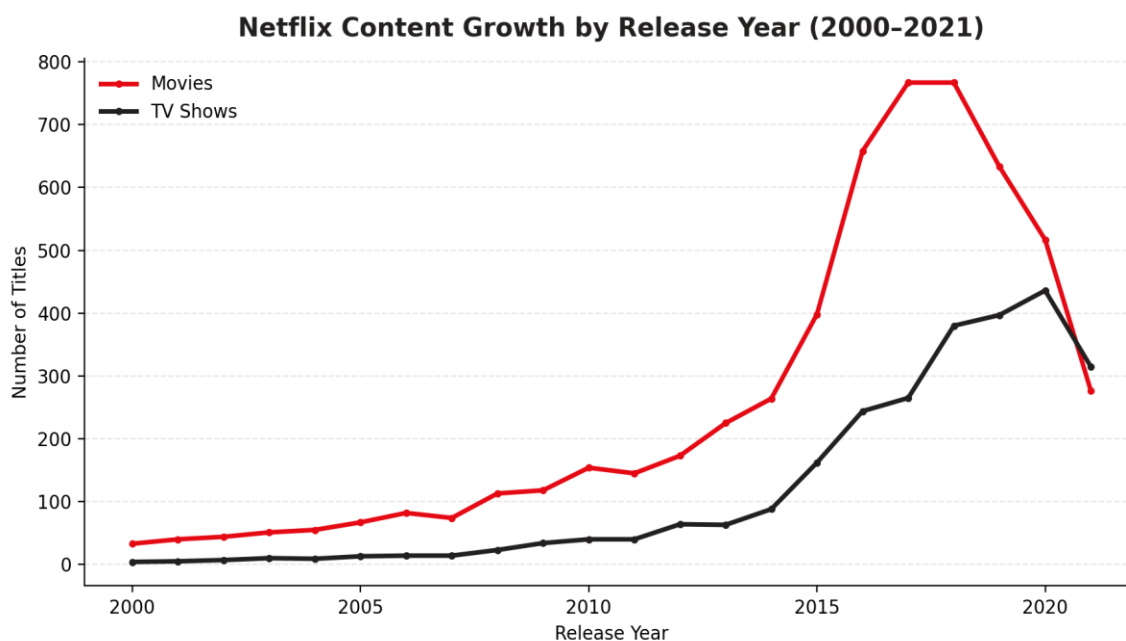


Figure 5: Movie vs. TV Show release volume by year (2000–2021).

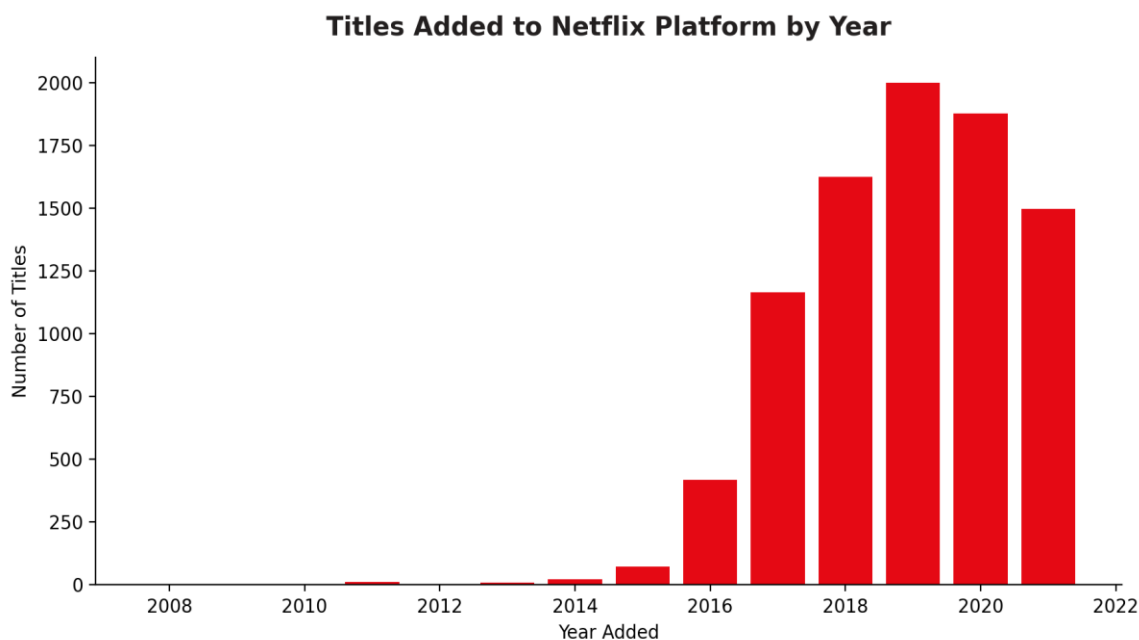


Figure 6: Titles added to the Netflix platform by calendar year.

Content additions slowed in 2020 and 2021, likely reflecting both a maturing catalog (diminishing returns on further volume growth) and pandemic-related production disruptions across the film and television industry.

8. Directors & Talent Analysis

Rajiv Chilaka leads all directors with 22 titles, primarily reflecting prolific animation/children's content production, followed by Jan Suter (21) and Raúl Campos (19), both closely associated with stand-up comedy specials. Martin Scorsese and Youssef Chahine (12 titles each) represent the more traditional feature-film segment of top contributors.

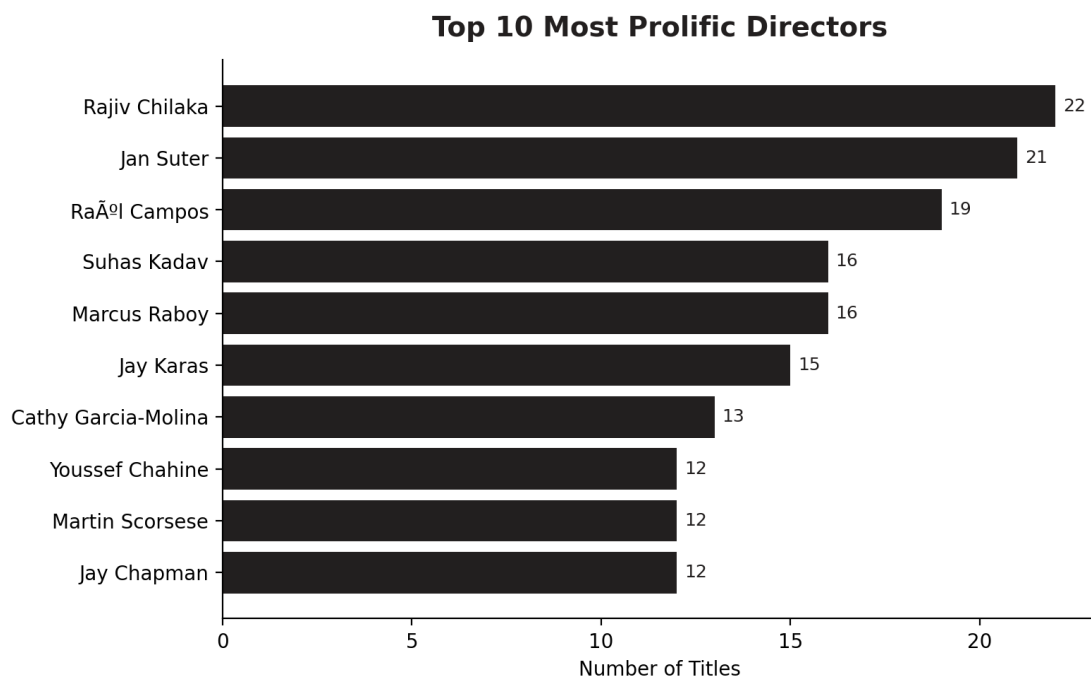


Figure 7: Directors with the highest number of catalog titles.

On the cast side, Anupam Kher (43 appearances), Shah Rukh Khan (35), and other prominent Indian actors feature heavily among the most frequently appearing talent — a further data point reinforcing the significance of Indian content within Netflix's global library.

9. Database-Level Analysis (PostgreSQL)

To validate and extend the Python-based findings, the cleaned dataset was loaded into PostgreSQL and queried using set-based SQL, including window functions for ranking and year-over-year trend analysis. Representative queries and their purpose are summarized below.

#	Business Question	SQL Technique Used
1	Top 10 directors by movie count	GROUP BY, COUNT, ORDER BY, LIMIT
2	Country ranking by title volume	UNNEST array split, RANK() OVER()
3	Longest movie in the catalog	CAST + ORDER BY DESC, LIMIT 1
4	Directors active in both formats	HAVING COUNT(DISTINCT type) = 2
5	Average movie duration by country	UNNEST, AVG, ROUND
6	Year-over-year growth in titles	LAG() OVER(), percentage calculation

The year-over-year growth query confirmed the pattern observed in Python: growth accelerated sharply between 2015 and 2018, then decelerated from 2019 onward as the catalog matured.

10. Interactive Dashboard (Power BI)

A companion Power BI dashboard (`netflix_dashboard.pbix`) was developed to translate these findings into an interactive, stakeholder-facing tool. The dashboard consolidates the key metrics from this report — content mix, country distribution, genre popularity, and yearly growth — into filterable visuals, allowing non-technical stakeholders to explore the data by year, country, or content type without writing code.

11. Key Insights Summary

- Movies outnumber TV shows nearly 2.3:1, indicating a historically film-first acquisition strategy.
- The United States and India are the two dominant content markets, together accounting for over half of all internationally-tagged production volume among top contributors.
- South Korea and Japan are the strongest TV-show-oriented markets, useful for targeted drama/anime content investment.
- TV-MA is the leading content rating, confirming a predominantly adult-audience catalog.
- Platform content additions peaked in 2019 and have since declined, suggesting a shift from volume growth to curation and originals strategy.
- International Movies and International TV Shows are the single largest genre categories, reflecting Netflix's global-content-first positioning rather than a domestic (US) focus.

12. Recommendations

- Continue investment in Indian and South Korean original productions, given their outsized representation and strong TV-show engagement patterns.
- Evaluate underrepresented markets (72 countries with fewer than 10 titles) for potential regional content-acquisition opportunities.
- Balance the movie-heavy catalog with further TV-series investment, particularly in genres showing decline (Dramas, Documentaries) to refresh audience interest.
- Use the PostgreSQL and Power BI layers built in this project as a recurring reporting pipeline, refreshed as new titles are added, to track catalog composition over time.

Appendix: Tools & Files

File	Purpose
<code>netflix_Analysis_1.ipynb</code>	EDA questions 1–10: content mix, countries, genres, ratings, directors
<code>Netflix_Analysis_2.ipynb</code>	EDA questions 11–20: trends, genre growth, country specialization
<code>Netflix_Analysis_3.ipynb</code>	EDA questions 21–29 + PostgreSQL data load via SQLAlchemy

File	Purpose
netflix_SQL.sql	10 core SQL business questions incl. window functions
netflix_dashboard.pbix	Power BI interactive dashboard
netflix_titles_0.xlsx	Source dataset (8,807 titles x 12 attributes)